

Comprehensive Project Report: Parameterized SPI Design & UVM Verification

1. Executive Summary

This project presents the complete RTL implementation and UVM verification of a parameterized Serial Peripheral Interface (SPI) IP core. The design features a flexible Master and Slave architecture with configurable clock divider, data width (8-32 bits), and full support for all 4 SPI modes. Verification was conducted using SystemVerilog and UVM 1.2 in AMD Vivado 2025.1, validating 100% full-duplex data integrity across corner cases and random tests.

2. Design Specifications Summary

• HDL: Verilog-2001 / SystemVerilog (IEEE 1800) • Target Architecture: AMD Xilinx Zynq UltraScale+ (xczu3eg-sbva484-2-e) • System Clock: 100 MHz • SPI Clock: 12.5 MHz (CLK_DIV=4) • Modes Supported: Mode 0 (0,0), Mode 1 (0,1), Mode 2 (1,0), Mode 3 (1,1) • Verification Methodology: UVM 1.2 (IEEE 1800.2)

3. Verification Results

The testbench executed 49 complex full-duplex transactions including boundary vectors (0x00, 0xFF, 0xAA, 0x55), walking bit patterns, all 4 clock modes, and 25 constrained-random packets.

Results: • Total Transactions: 49 • MOSI Matches: 49 / 49 (100%) • MISO Matches: 49 / 49 (100%) • UVM Errors: 0 • UVM FataIs: 0 • Status: ALL SPI TRANSACTIONS MATCHED SUCCESSFULLY!